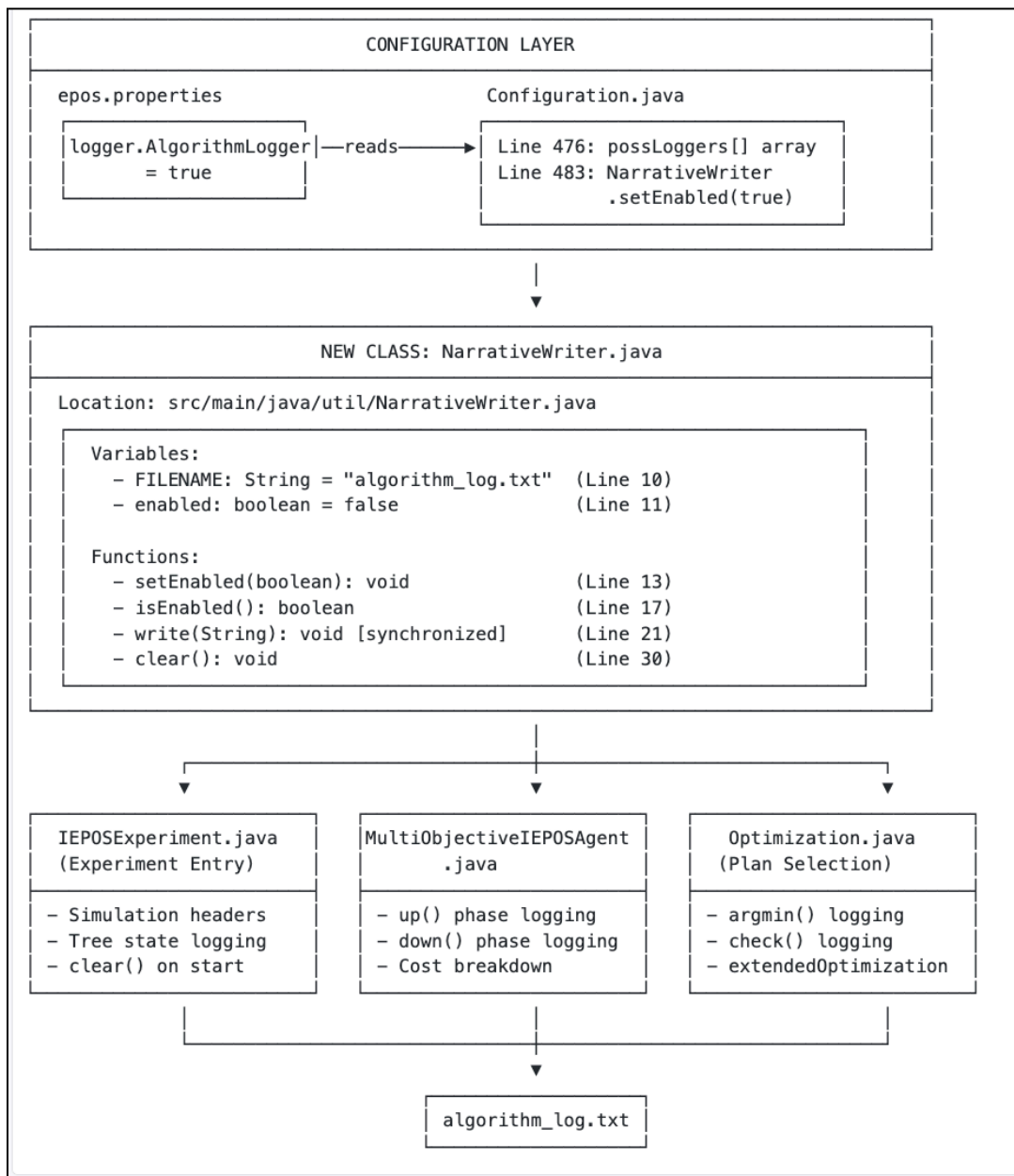


AlgorithmLogger – Documentation

1. Overview

The AlgorithmLogger is a custom logging feature that generates a human-readable, step-by-step narrative of the EPOS algorithm's execution flow. It writes to [algorithm_log.txt](#) and is controlled via the [logger.AlgorithmLogger](#) property in [epos.properties](#).

2. Architecture Diagram



3. New Class: NarrativeWriter

Property	Value
File	NarrativeWriter.java
Package	util
Type	New Class (Static Utility)

3.1 Variables

Variable	Type	Line	Description
FILENAME	String (static final)	10	Output file path: "algorithm_log.txt"
enabled	boolean (static)	11	Global toggle for the logger

3.2 Methods

Method	Signature	Line	Description
setEnabled	public static void setEnabled(boolean)	13-15	Sets the enabled flag
isEnabled	public static boolean isEnabled()	17-19	Returns the enabled flag
write	public static synchronized void write(String)	21-28	Appends message to log file (thread-safe)
clear	public static void clear()	30-38	Truncates the log file

4. Modifications to Existing Classes

4.1 Configuration.java

Property	Value
File	Configuration.java

Changes:

Line	Change Type	Code
476	Array modification	Added "logger.AlgorithmLogger" to possLoggers[]
482-483	New code block	util.NarrativeWriter.setEnabled(selectedLoggers.contains("logger.AlgorithmLogger"));

4.2 IEPOSExperiment.java

Property	Value
File	IEPOSExperiment.java

New Import:

Line	Code
35	import util.NarrativeWriter;

Changes:

Lines	Function/Context	Description
57-64	<code>runSimulation()</code> ▾	Logs simulation configuration header
140	<code>main()</code> ▾	NarrativeWriter.clear() - Clears log at experiment start
145	<code>main()</code> ▾	NarrativeWriter.write("\n=== Simulation " + ...) - Logs

		simulation number
234-252	New function logTreeSt... ▾	Logs tree structure at iteration end
268-285	New function printTree() ▾	Recursive helper for tree visualization
255-266	New function getFinger... ▾	Helper to extract agent ID from Finger

New Static Functions Added:

Function	Signature	Lines	Description
logTreeState	public static void logTreeState(int iteration)	234-252	Logs tree structure with selected plans
printTree	private static void printTree(TreeAgent, String, boolean)	268-285	Recursively prints tree
getFingerId	private static int getFingerId(Finger)	255-266	Extracts ID from Finger object

4.3 MultiObjectiveEPOSAgent.java

Property	Value
File	MultiObjectiveEPOSAgent.java

New Import:

Line	Code
9	import util.NarrativeWriter;

Changes:

Lines	Function	Description
296-309	New helper	formatFinger() Converts Finger to readable agent ID string

311-428	Override up()	Added detailed logging for BOTTOM-UP phase
442-563	Override down()	Added detailed logging for TOP-DOWN phase with full cost breakdown

New Helper Method:

Method	Line	Description
formatFinger(Finger f)	296-309	Converts a Finger object to "Agent X" format using Configuration.mapping

Logged Information in up() (Lines 311-428):

- Agent ID (mapped) and Iteration
- Leaf/Non-leaf status
- Children aggregate reception
- Selected Plan ID and cost
- Calculation breakdown (Children vectors, Sum, Result)

Logged Information in down() (Lines 442-563):

- Agent ID (mapped) and Iteration
- Global Response reception/calculation
- **Cost Analysis Breakdown:**
 1. Global Cost (Lines 489-501): Formula, calculation, weighted value
 2. Local Cost (Lines 503-517): Discomfort sum, average, weighted value
 3. Unfairness (Lines 519-533): Variance calculation ($E[X^2] - (E[X])^2$), std deviation, weighted value
 4. Total Cost (Lines 536-538): Sum of all weighted components

4.4 Optimization.java

Property	Value
File	Optimization.java

Changes:

Lines	Function	Description
207-245	argmin() (Proposal Selection)	Logs proposal evaluation with cost function details
277-393	check() (Plan Selection)	Logs plan evaluation and hard constraint checks
430-478	extendedOptimization()	Logs weighted cost calculation for each plan/proposal

Key Logging Points:

Lines	What is Logged
207-222	Proposal evaluation header, cost function names, number of combinations
236	Calls costFunction.logCostCalculation() for each proposal
245	Writes accumulated log via NarrativeWriter.write()
306	Calls costFunction.logCostCalculation() for each plan
312	Calls localCostFunction.logCostCalculation() for each plan
383-384	Hard constraint violation penalty
392	Writes plan selection log
456-460	Weighted cost calculation for $\alpha/\beta > 0$
478	Writes final optimization selection

4.5 ModifiableleposAgent.java

Property	Value
File	ModifiableleposAgent.java

New Import:

Line	Code
9	import util.NarrativeWriter;

Changes:

Lines	Function	Description
264-265	actionsToReorganize()	Logs reorganization events

4.6 Cost Function Interfaces & Implementations

New method [logCostCalculation\(\)](#) added to interfaces and implementations:

Interface: CostFunction.java

File	Lines	Default Method
CostFunction.java (Interface)	49-51	public default void logCostCalculation(V value, StringBuilder sb)

Interface: PlanCostFunction.java

File	Lines	Default Method
PlanCostFunction.java (Interface)	34-36	public default void logCostCalculation(Plan<V> plan, StringBuilder sb)

Implementations:

File	Lines	Description
VarCostFunction.java	37-60	Logs variance calculation (mean, sum of squares)

CustomCostFunction.java	42-65	Logs dot product calculation
PlanDiscomfortFunction.java	28-31	Logs discomfort score
PlanPreferenceFunction.java	25-28	Logs preference index

5. Summary Table of All Changes

File	Lines Modified/Added	Type	Purpose
util/NarrativeWriter.java	1-39	NEW CLASS ▾	Core logging ut... ▾
config/Configuration.java	476, 482-483	Modified ▾	Enable/disable ... ▾
experiment/IEPOSExperiment.java	35, 57-64, 140, 145, 234-285	Modified + New... ▾	Simulation hea... ▾
agent/MultiObjectiveEPO SAgent.java	9, 296-309, 311-428, 442-563	Modified ▾	UP/DOWN pha... ▾
agent/Optimization.java	207-245, 277-393, 430-478	Modified ▾	Plan/Proposal ... ▾
agent/ModifiableleposAgent.java	9, 264-265	Modified ▾	Reorganization... ▾
func/CostFunction.java	49-51	NEW METHOD ▾	Interface for log... ▾
func/PlanCostFunction.java	34-36	NEW METHOD ▾	Interface for log... ▾
func/VarCostFunction.java	37-60	NEW METHOD ▾	VAR cost logging ▾
func/CustomCostFunction.java	42-65	NEW METHOD ▾	Dot product log... ▾
func/PlanDiscomfortFunction.java	28-31	NEW METHOD ▾	Discomfort logg... ▾

6. Algorithm Flow with Logging



7. Configuration

To enable/disable in epos.properties:

```
logger.AlgorithmLogger = true  ## Enable detailed algorithm logging  
logger.AlgorithmLogger = false ## Disable (default)
```